

Package ‘SFPLSR’

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Type Package

Title Spatial Functional Partial Least Squares Regression

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Description This package implements the Spatial Functional Partial Least Squares (SFPLS) framework for spatial autoregressive scalar-on-function regression modeling, integrating spatial dependence into the dimension reduction stage.

License GPL-3

Encoding UTF-8

Roxygen list(markdown = TRUE)

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Depends R (>= 3.5)

LazyData true

Imports expm,
fda,
fda.usc,
MASS,
stats

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predict_sfplsr	<i>Predict Method for Spatial FPLS Regression</i>
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Description

Predict Method for Spatial FPLS Regression

Usage

```
predict_sfplsr(object, xnew, wnew)
```

Arguments

object	An object obtained from the splsr function.
xnew	A numeric matrix of new functional predictors (new observations).
wnew	A square spatial weights matrix for the new observations or test validation block.

Value

A numeric vector of predicted values for the scalar response.

sfplsr	<i>Spatial Functional Partial Least Squares Regression</i>
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Description

Spatial Functional Partial Least Squares Regression

Usage

```
sfplsr(
  y,
  x,
  h,
  nbasis = NULL,
  gpx = NULL,
  wei_mat,
  rho_grid = seq(-0.95, 0.95, by = 0.01),
  criterion = c("train_mse", "bic", "aic"),
  scv = FALSE,
  block_ids = NULL,
  verbose = TRUE
)
```

Arguments

<code>y</code>	A numeric vector of the scalar response variable.
<code>x</code>	A numeric matrix of functional explanatory variable.
<code>h</code>	An integer or vector specifying the number of latent components.
<code>nbasis</code>	An integer or vector specifying the number of basis functions.
<code>gpx</code>	A numeric vector defining the evaluation grid points for <code>x</code> .
<code>wei_mat</code>	A square spatial weights matrix.
<code>rho_grid</code>	A numeric vector defining the search grid for spatial parameter ρ .
<code>criterion</code>	Optimization metric: "train_mse", "bic", or "aic".
<code>scv</code>	Logical; if TRUE, performs Spatial Cross-Validation for <code>nbasis</code> and <code>h</code> .
<code>block_ids</code>	A vector indicating the spatial block assignment for each observation (required if <code>scv</code> = TRUE).
<code>verbose</code>	Logical; if TRUE, prints progress and tuning results.

spatial_data_generation

Generate Spatial Functional Data

Description

Generate Spatial Functional Data

Usage

```
spatial_data_generation(n, j, rho, sig.e)
```

Arguments

<code>n</code>	An integer specifying the sample size (number of spatial locations).
<code>j</code>	An integer specifying the number of grid points for the functional predictor.
<code>rho</code>	A numeric value representing the spatial autoregressive parameter.
<code>sig.e</code>	A numeric value specifying the standard deviation of the error terms.

Value

A list containing the generated spatial functional dataset components.

UK_weather_data	<i>UK Weather Data for Spatial Functional Partial Least Squares Regression</i>
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Description

A dataset containing solar radiation, temperature, and station info from UK weather stations used in the SFPLS framework.

Usage

UK_weather_data

Format

A named list containing the weather station variables.

Source

NASA POWER Dataset

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